King Saud University
Muzahimya branch
College of Applied computer science



Subject: Network Protocols

ID Course: cos372

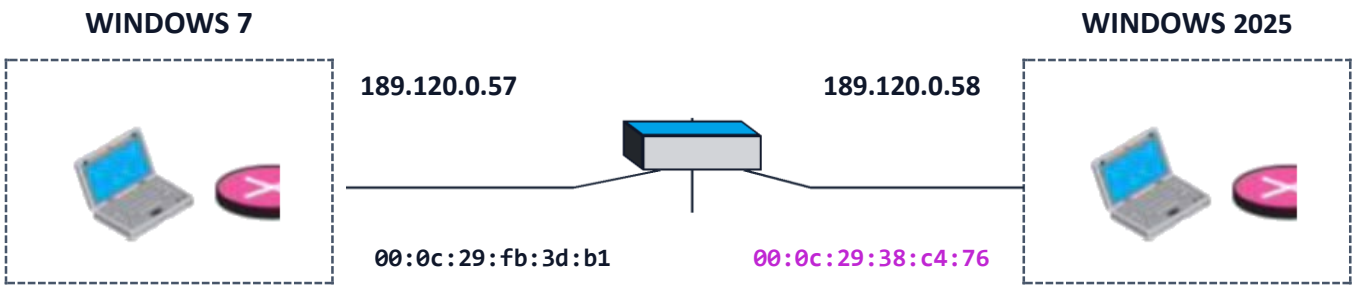
Term: 2026fall

Report title: ICMP + ARP

Project No: 1

Prof: Wail S.Elkilani

Name: Fahad Awad



General Formula for ARP Message (28 Bytes)

Hardware Type (2 Bytes / 16 bits)		Protocol Type (2 Bytes / 16 bits)
Hardware Length (1 Byte / 8 bits)	Protocol Length (1 Byte / 8 bits)	Operation Code (Opcode) (2 Bytes / 16 bits)
Sender Hardware Address (Sender MAC) (6 Bytes / 48 bits)		
Sender Protocol Address (Sender IP) (4 Bytes / 32 bits)		
Target Hardware Address (Target MAC) (6 Bytes / 48 bits)		
Target Protocol Address (Target IP) (4 Bytes / 32 bits)		

Preamble & SFD (8 Bytes)	Destination MAC (6 Bytes)	Source MAC (6 Bytes)	Type: 0x0806 (2 Bytes)	Data (ARP Message) (28 Bytes)	Padding (18 Bytes)	CRC / FCS (4 Bytes)

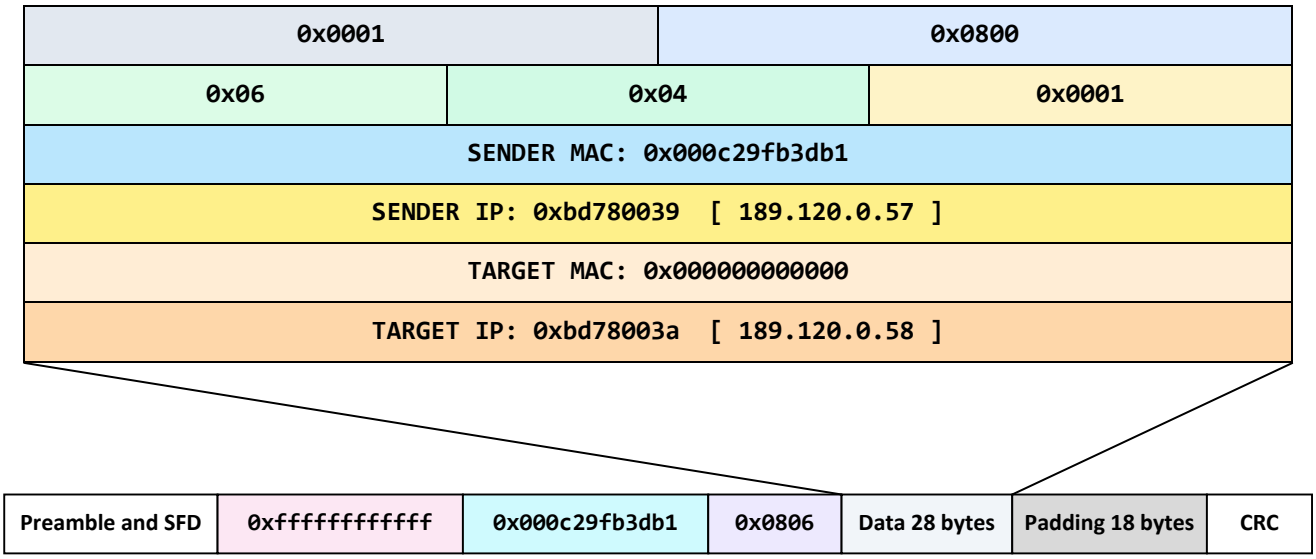
General Formula for ICMP Echo Request & Reply Message

Type: 8 or 0 (1 Byte / 8 bits)	Code: 0 (1 Byte / 8 bits)	Checksum (2 Bytes / 16 bits)
Identifier (2 Bytes / 16 bits)		Sequence number (2 Bytes / 16 bits)
Optional data Sent by the request message; repeated by the reply message		

IP Header (20 Bytes)	IP Data (ICMP Message) (8 Bytes + Optional data)

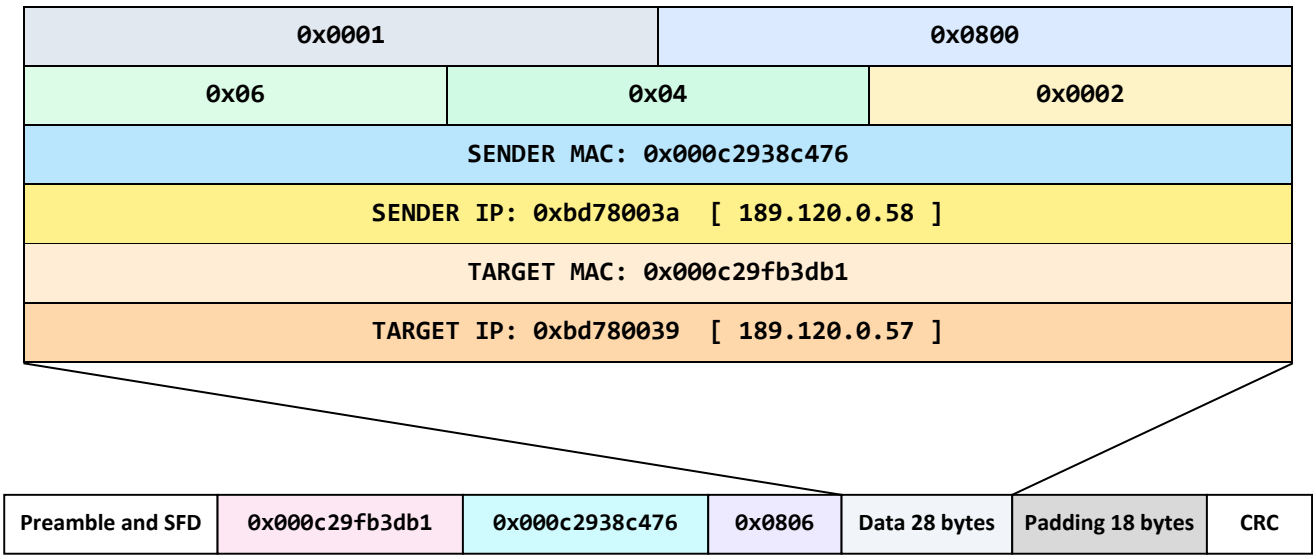
Preamble & SFD (8 Bytes)	Destination MAC (6 Bytes)	Source MAC (6 Bytes)	Type: 0x0800 (2 Bytes)	Data (IP Datagram) (up to 1500 Bytes)	Padding (0 to 18 Bytes)	CRC / FCS (4 Bytes)

ARP Request



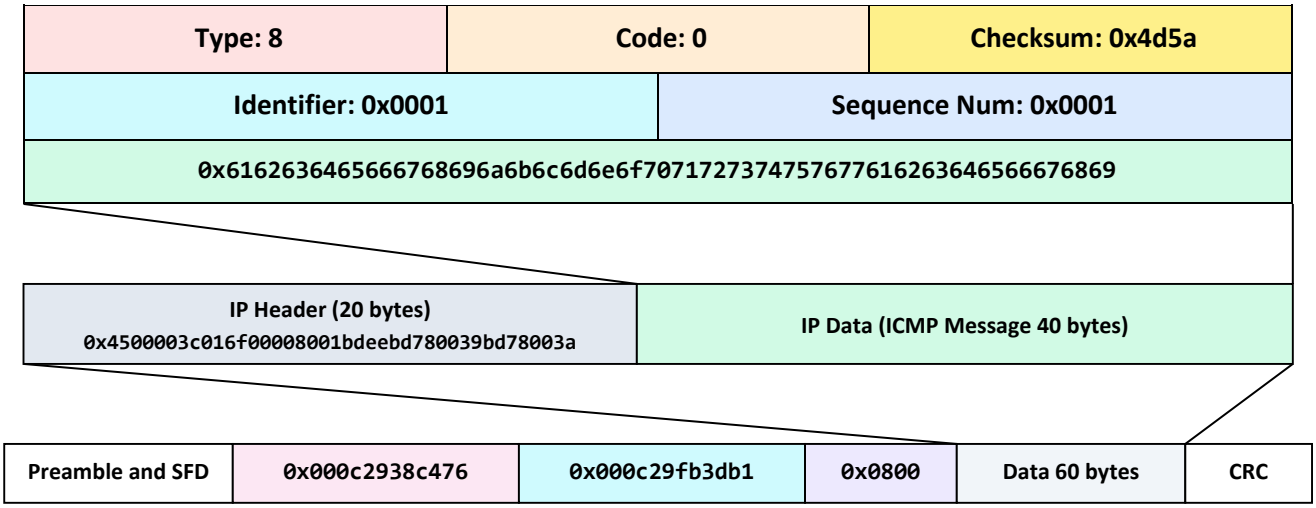

```
ff ff ff ff ff ff 00 0c 29 fb 3d b1 08 06 00 01
08 00 06 04 00 01 00 0c 29 fb 3d b1 bd 78 00 39
00 00 00 00 00 00 bd 78 00 3a 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00
```

ARP Reply



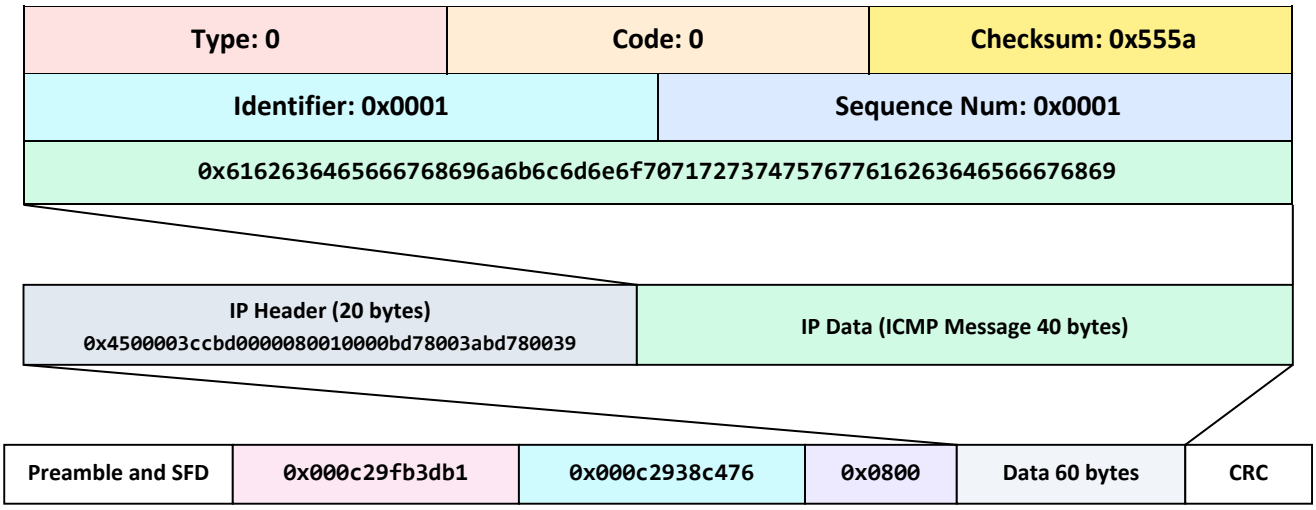

```
00 0c 29 fb 3d b1 00 0c 29 38 c4 76 08 06 00 01
08 00 06 04 00 02 00 0c 29 38 c4 76 bd 78 00 3a
00 0c 29 fb 3d b1 bd 78 00 39
```

ICMP REQUEST




```
        1     2     3          4          5
00 3a   08    00    4d 5a    00 01    00 01    61 62 63 64 65 66
67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
77 61 62 63 64 65 66 67 68 69
```

ICMP REPLY




```
        1     2     3          4          5
00 39   00    00    55 5a    00 01    00 01    61 62 63 64 65 66
67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76
77 61 62 63 64 65 66 67 68 69
```